

Lifting 2D Keypoints to 3D Human Pose Estimation

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Abstract

3D human pose estimation involves estimating the articulated 3D joint locations of a human body from an image or a video. However, most image and video data are in 2D format, which brings up the difficulty of solving the depth ambiguity and occlusion problem that comes with lifting 2D pose to 3D. In addition, the limited availability of in-the-wild datasets with 3D ground truth data and self-occlusion results in ambiguous multiple possible 3D human poses. Research has grown from using convolutional architecture to using transformers and much more recently spatial-temporal architectures (PoseFormer) [25] and multi-hypothesis architecture (MHFormer) [13]. Building off from the most recent and accurate architecture, MHFormer, we propose an alternative method of improving the architecture that might produce an improvement in accuracy. We alter the connection between two critical parts of the MHFormer (Self Hypothesis Refinement (SHR) and Cross Hypotheses Interaction (CHI) - explained in details in methods) in order to increase the interaction between the multiple hypothesis. Our current result shows a slightly lower performance than the original MHFormer method but our experiments reveals that a combination of MHFormer and PoseFormer would perform generalize better in both conditions of occluding and non-occluding situations.

1. Introduction

Vast majority of existing depictions of humans are two dimensional, ranging from video footage, images or paintings and these representations have traditionally played an important role in conveying facts, and ideas. However, this method of transmitting information is only useful for humans who can interpret complex spatial and temporal arrangements in the presence of depth ambiguities. For the applications like virtual & augmented reality, and autonomous driving, giving this spatial and temporal reasoning power to machines is crucial. To elaborate, today ma-

jority of self-driving car accidents are caused by “robotic” driving, where the self-driving vehicle conducts an allowed (abrupt) but unexpected stop, and a human driver crashes (from the rear or sides) into the self-driving car. With real-time human pose detection and tracking, the computers are able to understand and predict pedestrian behavior much better – allowing more natural driving and enhancing the road safety.

There are two prominent approaches to tackle 3D human pose estimation: Direct 3D pose estimation (end-to-end learning) and lifting 2D to 3D human pose. Direct 3D human pose estimation is solved using two methods, namely, detection based [14] and regression based [15]. However, the direct method suffers from the lack of large 3D human pose dataset and obtaining 3D ground truth dataset is quite expensive. Another promising approach is lifting 2D to 3D human pose. The approach includes two procedures that first translate the image into 2D human pose and then convert the 2D human pose to 3D human pose. Existing methods for translating images to 2D human pose include OpenPose [1] that outputs 2D body part locations in cartesian coordinates with the corresponding confidence, and EpipolarPose [12] that outputs body part locations in epipolar coordinates. Both methods have proven high accuracy in predicting the 2D human poses, and the remaining problem at interest is the transition from 2D human poses to 3D human poses with high accuracy. For our project, we use video as an input for our project and perform 2D key-point detection using an off-the-shelf Cascaded Pyramid Network [4] and output 3D keypoint detections.

A fundamental problem in computer vision is to understand the 3D geometry of a scene captured in 2D. This involves describing the scene in terms of the objects present in it (i.e. object detection and classification) and predicting the relative rigid transformation between the camera and each object (i.e. object pose estimation) [25]. In human pose estimation, even if 2D human poses could be extracted from images with high accuracy, the 3D pose estimation is difficult. The difficulty comes from the non-trivial mapping

between 2D poses and 3D poses due to the depth ambiguity and occlusion of the input image [25]. Recently, transformers have been used in natural language processing due to its scalability, efficiency and strong modeling capabilities. Due to the self-attention mechanism of the transformer, global correlations across long input sequences can be distinctly captured and this makes it a suitable architecture for sequence data problems and therefore naturally extendable to 3D HPE. However, recent works [5] show that transformers require specific designs to achieve comparable performance with CNN counterparts for vision tasks. Specifically, they often require either extremely large scale training datasets, or enhanced data augmentation and regularization [21] if applied to smaller datasets. Based on development on the usage of transformer network, multiple architecture have been developed with the recent one being multiple hypothesis transformer which is the state of art.

2. Related Work

The 2D to 3D pose estimation method involves a two-step process: (1) localize 2D joints with a 2D pose estimator on each view, and (2) lift the 2D joints from multi-view images to the 3D position and it is more preferred among the state-of-the-art methods in this active area of research. In 2D-to-3D lifting approach, research carried out have heavily relied on OpenPose [1], CPN [4] and AlphaPose [6] for the 2D pose detectors. For the translation between 2D human pose and 3D human pose, Martinez [16] has proposed a simple baseline architecture composed of two sequential blocks with a linear layer followed by batch normalization, ReLU activation and dropout. Pavllo *et al.* has proposed a temporal convolution structure that takes in a sequence of frames to predict the 3D human pose [19]. Hossain and Little [8] proposed a recurrent neural network using Long Short-Term Memory (LSTM) cells to exploit temporal information in the input sequence.

In recent times, transformer architecture [22] has been proven to play a key part of computer vision task including object detection [26], image classification [3] and semantic segmentation [8] and more closely to human pose estimation task and have proven better results. Carion *et al.* [2] presented DETection TRansformer (DETR) for object detection and panoptic segmentation. Dosovitskiy *et al.* [5] proposed a pure transformer architecture, Vision Transformer (ViT), which achieves state-of-the-art performance on image classification. However, ViT was trained on large-scale data sets ImageNet-21k and JFT-300M that requires huge computational resources. Then, a data-efficient image transformer (DeiT) [21] was proposed which builds upon the ViT with knowledge distillation.

One drawback of using transformer model in 3D human pose estimation is that the estimation from monocular video has lower accuracy in the occluding situations, and

hence generating a single solution (3D HPE) is sub-optimal. Works on multiple hypothesis for the inverse problem of 3D HPE achieve better performance gains. One-to-many mapping performed by [11] generates multiple 3D pose candidates consistent with 2D key-points. [23] modeled the posterior distribution of 3D hypotheses using normalized flows. On the other hand, MHFormer [13] learn one-to-many mapping first and then deploys many-to-one mapping which allows effective modeling of different features corresponding to various hypotheses in order to improve the representation ability.

3. Methods

3.1. Multi-hypothesis Transformer

This work uses the transformer-based architecture which performs well in long-range dependency modeling. The basic components of the transformer are the multi-head self-attention (MSA) and a multi-layer perception (MLP).

MSA. In the MSA, the inputs are $x \in \mathbb{R}^n$, and the keys are $K \in \mathbb{R}^{n \times d}$, where n is the sequence length, and d is the embedding dimension. The scaled-dot product attention can be computed by:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (1)$$

MLP. The MLP consist of two linear layers along with a non-linear activation:

$$MLP(x) = \sigma(xW_1 + b_1)W_2 + b_2 \quad (2)$$

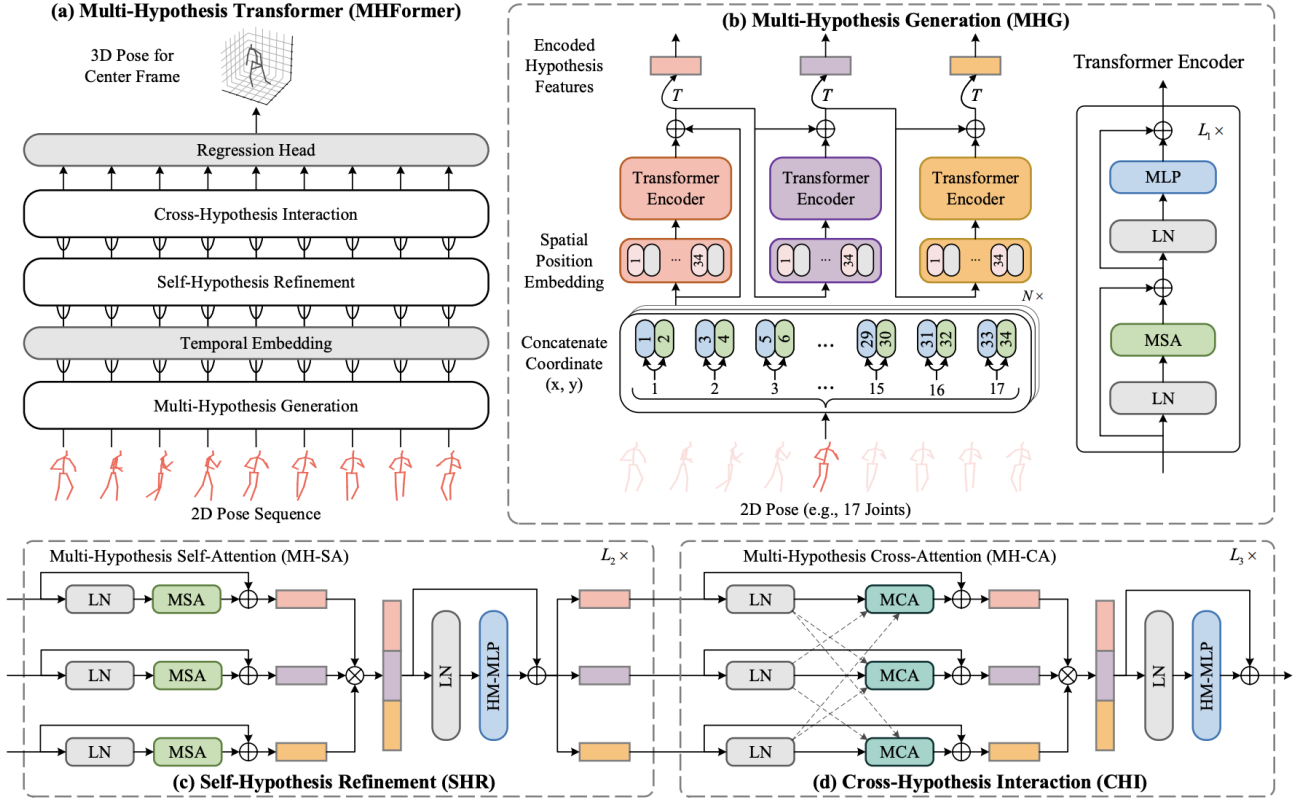
where σ denotes the GELU activation function, $W_1 \in \mathbb{R}^{d \times d_m}$ and $W_2 \in \mathbb{R}^{d_m \times d}$ are the weights of the two linear layers respectively, and $b_1 \in \mathbb{R}^{d_m}$ and $b_2 \in \mathbb{R}^d$ are the bias terms. The overall architecture of the Multi-hypothesis Transformer is showed in the Fig 1.

3.2. Multi-Hypothesis Generation(MHG)

The MHFormer paper tackles the inverse problem by designing a cascaded transformer-based architecture to generate multiple features in different depths of latent space [13]. MHG model is introduced: suppose there are M different hypotheses and L1 layers in the MHG, it takes a sequence of 2D poses $X \in \mathbb{R}^{N \times J \times 2}$ with N video frames and J body joints as input and outputs multiple hypotheses $[X_{L1}^1, X_{L1}^2, \dots, X_{L1}^M]$, where $X_{L1}^m \in \mathbb{R}^{(J \cdot 2) \times N}$ is the m-th hypothesis.

We concatenate the (x, y) coordinates of joints for each frame to $\bar{X} \in \mathbb{R}^{(J \cdot 2) \times N}$, retain their spatial information of joints via a learnable spatial position embedding $E_{SP_{os}}^m \in$

Figure 1. MHFormer model architecture [13]



$\mathbb{R}^{(J-2) \times N}$, and feed the embedded features into the encoders of the MHG. Skip residual connection as written in the paper [13] is also applied between the original input and output features.

3.3. Temporal Embedding

One limitation with the original architecture is the weak initial multi-hypothesis feature generation and this limitation is addressed by building relationships across hypothesis features and capturing temporal dependencies in the temporal domain with two carefully designed modules: an SHR module followed by a CHI module.

In order to exploit temporal information, we should first convert the spatial domain into the temporal domain. The encoded hypothesis features X_{L1}^m of each frame are embedded to the high-dimensional features $Z^m \in \mathbb{R}^{N \times C}$ using a transposition operation and linear embedding where C is the embedding dimension.

3.4. Self-Hypothesis Refinement

Features of each hypothesis are refined via Self-Hypothesis Refinement (SHR) which consists of two main blocks: multiple hypothesis self-attention (MH-SA) and hypothesis-mixing MLP.

Multiple Hypothesis Self-Attention (MH-SA) is used to capture the dependencies between the features of single hypothesis. The architecture of this block is similar to the multi-head self-attention (MSA).

Multi-head Self Attention (MSA) is used to generate several different attention heads that could attend to different parts of the single hypothesis features separately. Mathematically, a single input tensor $x \in \mathbb{R}^{n \times d}$ is linearly mapped to query $Q_x^i \in \mathbb{R}^{n \times d}$, key $K_x^i \in \mathbb{R}^{n \times d}$ and value $V_z^i \in \mathbb{R}^{n \times d}$, where i corresponds to different attention head. The scaled dot-product attention for MSA at the i^{th} head is computed as:

$$Attention(Q^i, K^i, V^i) = Softmax\left(\frac{Q^i K^{iT}}{\sqrt{d}}\right) V^i \quad (3)$$

At the end of one MSA, multiple heads are then concatenated and fed into a linear layer. For the m^{th} hypothesis passing into the $(l-1)^{th}$ SHR layer, Z_{l-1}^m , the output of the block is computed as:

$$Z_l^m = Z_{l-1}^m + MSA^m(LN(Z_{l-1}^m)) \quad (4)$$

Hypothesis-Mixing MLP is used to facilitate information exchange across hypotheses. Features are concatenated and

fed into hypothesis mixing MLP to merge themselves. MH-CA are fed into hypothesis-mixing MLP and from the output of the last CHI layer partition operation is not done so as features of all the hypotheses can be aggregated to synthesize a single hypothesis representation $Z_{L_3} \in R^{N \times (C \cdot M)}$

$$Z'_l = \text{Concat}(Z'^{1}_l, \dots, Z'^{M}_l) \in R^{N \times (C \cdot M)} \quad (5)$$

$$\text{Concat}(Z'^{1}_l, \dots, Z'^{M}_l) = Z'_l + \text{HM} - \text{MLP}(\text{LN}(Z'_l)) \quad (6)$$

3.5. Cross-Hypothesis Interaction

Interaction between multiple hypothesis is ensured via Cross-Hypothesis Interaction (CHI) which consists of two main blocks: multiple hypothesis cross-attention (MH-CA) and hypothesis-mixing MLP.

Multiple Hypothesis Cross-Attention (MH-CA) is used to facilitate the interaction between different hypothesis. On the other hand, multiple hypothesis self attention (MH-SA) does not facilitate interaction across different hypothesis and hence reduces interaction between different hypothesis. MH-CA comprises of numerous parallel multi-head cross attention (MCA) elements.

Multi-Head Cross Attention (MCA) is used to find the correlation among cross-hypothesis and it fairly resembles the structure of MSA. However, as compared to common configuration of MCA, there is an exception of using different inputs (i.e. different hypothesis) instead of normal configuration which uses the same input between keys and values. More formally, MCA can be mathematically explained as follows for three tensors, x , y , and z . The inputs $x \in R^{n \times d}$, $y \in R^{n \times d}$, and $z \in R^{n \times d}$ are linearly mapped to queries $Q_x \in R^{n \times d}$, keys $K_y \in R^{n \times d}$ and values $V_z \in R^{n \times d}$ respectively. The scaled dot-product attention in the MCA can be computed by:

$$\text{Attention}_{\text{cross}}(Q_x, K_y, V_z) = \text{Softmax}\left(\frac{Q_x K_y^T}{\sqrt{d}}\right) V_z \quad (7)$$

The common configuration of MCA uses the same input between keys and values i.e., the inputs $x \neq y = z$. For computation efficiency and reducing the number of blocks from 6 to 3, MHFormer uses different inputs (individual hypothesis Z^m), i.e., inputs $x \neq y \neq z$ are fed into MH-CA.

$$Z'^m_l = Z^m_{l-1} + \text{MCA}^m(\text{LN}(Z^m_{l-1}), \text{LN}(Z^m_{l-2}), \text{LN}(Z^m_{l-1})) \quad (8)$$

where CHI layers $l \in [1, 2, \dots, L_3]$, $Z^m_0 = \tilde{Z}^m_{L_2}$, m_1 , and m_2 are two hypotheses, and $\text{MCA}(Q, K, V)$ denotes the function of the MCA.

3.6. Regression Head

Output from the CHI layer Z_{L_3} is passed through linear transformation layer to perform regression to produce 3D pose sequence $\hat{X} \in R^{N \times J \times 3}$. Finally, 3D pose of the center frame $\hat{X} \in R^{J \times 3}$ is selected from $\hat{X}$ for our final prediction.

3.7. Loss Function

Mean squared error (MSE) loss is applied to minimize the error between the estimated pose ($\hat{X}_i^n$) and ground truth (Y_i^n) poses for i^{th} joint at frame n , the loss function uses L2 norm.

$$\mathcal{L} = \sum_{n=1}^N \sum_{i=1}^J \|Y_i^n - \hat{X}_i^n\|_2 \quad (9)$$

4. Dataset

The Human3.6M dataset [10] provides accurate capture of pixel-level 24 body parts labels from 3.6M images under different scenarios and consistent actors, synchronized under high resolution 50 Hz 4-point video camera. Human3.6M is, to the best of our knowledge, currently the largest publicly available datasets for human 3D pose estimation. The dataset consists of 15 everyday activities such as walking, eating, sitting, making a phone call and engaging in a discussion from 15 different subjects, as shown in Fig. 2

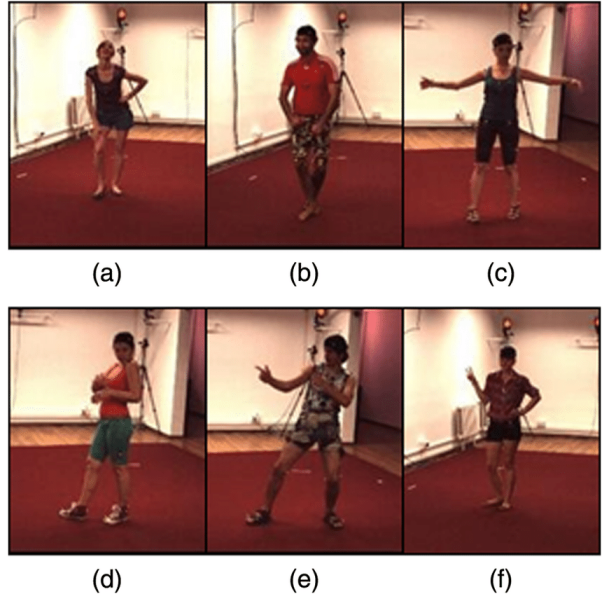


Figure 2. Human 3.6M dataset sample images [10]

2D joint locations and 3D ground truth positions are available, as well as projection (camera) parameters and body proportions for all the actors. Both MHFormer and

PoseFormer in the paper are training on this dataset with subject 1, 5, 6, 7, 8, and validating with subject 9, 11 [13] [25]. Altogether there are 3,119,552 frames for training, and 543,360 frames for validation.

4.1. Pre-Processing

For implementing the baseline method, we obtain 2D detections using cascaded pyramid network (CPN) [4]. As proposed by Martinez [16], a square of size 440×440 pixels around the center of the detected person (center provided by the Human3.6M dataset) is extracted as the input of the 2D detector for pre-processing.

5. Experiments

5.1. Baseline Model

Fig 3 depicts the basic building blocks of the baseline method [16]. This model consists of multi-layer neural network with batch normalization [9], dropout [20], and Rectified Linear Units (ReLU) [17] activation function and residual connections [7].

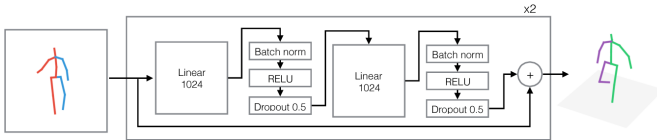


Figure 3. Baseline Network Architecture [16]

We train the baseline model using subjects 1, 5, 6, 7, 8 of Human 3.6M dataset and evaluate using subjects 9 and 11. We report errors between prediction and ground truth values across all joints and cameras, after alignment of the root in millimeters (mm). We trained the model for 15 epochs as shown in Table 4.

5.2. Transformer Models

We considered working with two recent transformer models, namely MHFormer [13], and PoseFormer [25]. We elucidate the two models below:

5.2.1 MHFormer

We follow the architecture of the MHFormer paper as shown in Fig 1 above and experiment on its architecture by tweaking the arrangement of SHR and CHI blocks. In the MHFormer architecture, there are two SHR blocks followed by one CHI block ($SHR \rightarrow SHR \rightarrow CHI$) (as shown in fig 4 (a)). We experiment with the above arrangement by changing the number and arrangement of these blocks to have the following architecture: $SHR \rightarrow CHI \rightarrow SHR \rightarrow CHI$ (as shown in fig 4 (b)). This is then fed

into the regression head and the rest of the architecture remains the same. A demonstration of the new architecture is shown below:

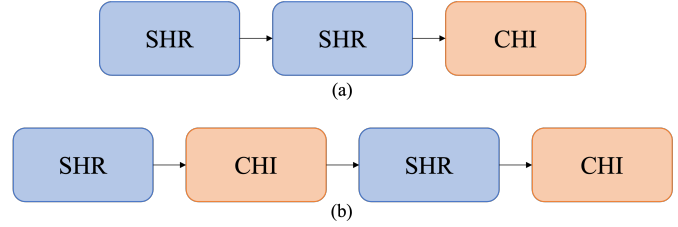


Figure 4. MHFormer model experiments

Motivation for model experiment: Original MHFormer method utilized SHR for finding the self attention with each of the generated hypotheses, and CHI for obtaining cross attentions across multiple hypotheses. In Fig 4 (a), the interaction between different hypotheses takes place during CHI (one time in their original architecture), which produces the state of the art result. Our motivation to propose the change shown in Fig 4 (b) is to increase the interaction (cross-attention) between different hypotheses in generating higher degree of interaction between multiple hypotheses which might improve the validation accuracy.

While proposing the above experiment, we have increased the number of layers from 3 (two SHR and one CHI) to 4 (two of each SHR and CHI). Increasing the total number of layers will improve the model expressivity (keeping the same number of hypothesis), which is similar to a deeper neural network.

5.2.2 PoseFormer

The PoseFormer architecture relies on spatial-temporal transformers which consists of three modules: a spatial transformer module for extracting features with considering joints correlations of each individual skeleton, a temporal transformer module for learning global dependencies of entire sequence and a regression head module that regresses the final 3D pose of the center frame. The major difference between the PoseFormer and MHFormer is that the MHFormer generates multiple feasible solutions (hypothesis) which are concatenated to form one refined solution while PoseFormer generates only one feasible solution.

5.3. Model Training

Since we are carrying out changes to the model architecture, we carried out training of the above three architecture from scratch using Pytorch Framework on AWS deep learning AMI 18.04 machine using p2.xlarge and g4dn.xlarge instances. Since each epoch takes approximately 20 hours for

training and 3 hours for validation, we chose to run each of the architecture for 2 epochs and a subset of 80%. Subjects 1, 5, 6 of Human 3.6M dataset are selected as the training set, and subject 9 is used for validation. The default settings (same as the published paper) are applied, with the initial learning rate being 0.0004 for the PoseFormer and 0.001 for the MHFormer and alternative MHformer, all with a learning rate decay of 0.95 applied after each epoch. We perform 2D pose detection using Cascaded Pyramid Network (CPN) for Human3.6M. The results from the PoseFormer, MHFormer and our new architecture (MHFormer alternate, Fig 4 (b)) are compared as shown in the result section.

6. Results

6.1. Evaluation Metrics

The performance of the model depends on the estimation accuracy of the joint position. For quantitative metrics of performance, we report and evaluate the model using these two measures: Mean Per Joint Position Error (MPJPE) and Procrustes analysis (P-MPJPE) [24]. MPJPE measures the mean Euclidean distance (the average L2 norm) of the difference between the estimated position and ground truth of each joint in millimeters; P-MPJPE is evaluated after aligning the predicted 3D pose with the ground truth using translation, rotation, and scale.

$$MPJPE(f, \mathcal{S}) = \frac{1}{N_S} \sum_{n=1}^{N_S} \|P_{f, \mathcal{S}}^{(f)}(i) - P_{gt, \mathcal{S}}^{(f)}(i)\|_2 \quad (10)$$

where f denotes the frame index and $\mathcal{S}$ denotes the corresponding skeleton. $P_{f, \mathcal{S}}^{(f)}(i)$ is the estimated position of joint i and $P_{gt, \mathcal{S}}^{(f)}(i)$ is the corresponding ground truth.

6.2. Quantitative Results

We trained our baseline method [16], for 15 epochs as our project milestone and obtained validation performance for subject 9 and 11 as reported in the Table 4. This baseline method used stacked hourglass method for 2D keypoint detection [18]. We observe the validation training loss decreases from 0.119 to 0.0491 as the number of epochs increases. This represents a 58.7 % improvement from the beginning epoch.

To explore state of the art methods, we experiment with transformer based model. We report the quantitative results for transformer based models in Table 1 and 2: MHFormer original (MHF-Original) (Fig 4 (a)), MHFormer alternate (MHF-Alt) (Fig 4 (b)) and PoseFormer (PF-Original) trained for 2 epochs of train/validation.

From table 1 and 2, we find that our method generally performs second best for actions like Discussion, Greetings, Phoning, Sitting, SittingDown, Waiting, Walking actions. Reason for such performance can be attributed to increased

Table 1. MPJPE results of Transformer models, where MHF - MHFormer, MHF-Alt - MHFormer Alternative, PF-Original - PoseFormer Original Architecture. Best in **bold**, second best underlined

Actions	MHF-Original	MHF-Alt	PF-Original
Directions	48.58	52.48	<u>48.77</u>
Discussion	49.99	<u>51.35</u>	59.05
Eating	<u>45.86</u>	48.25	44.98
Greeting	51.16	<u>54.02</u>	54.41
Phoning	55.15	<u>58.82</u>	61.39
Photo	<u>65.45</u>	70.09	61.14
Posing	<u>49.87</u>	51.47	47.5
Purchases	<u>52.82</u>	52.9	50.25
Sitting	69.49	73.4	76.21
SittingDown	72.22	<u>73.58</u>	85.07
Smoking	54.34	56.55	<u>56.36</u>
Waiting	52.36	<u>51.64</u>	51.02
WalkDog	<u>54.6</u>	55.55	54.39
Walking	<u>44.71</u>	47.09	39.85
WalkTogether	<u>45.59</u>	50.58	43.98
Average	54.15	56.52	<u>55.59</u>

Table 2. P-MPJPE results of Transformer models, where MHF - MHFormer, MHF-Alt - MHFormer Alternative, PF-Original - PoseFormer Original Architecture. Best in **bold**, second best underlined

Actions	MHF-Original	MHF-Alt	PF-Original
Directions	36.32	<u>36.58</u>	37.25
Discussion	39.35	40.63	42.82
Eating	35.94	36.78	35.72
Greeting	39.46	<u>40.83</u>	42.03
Phoning	42.67	<u>44.52</u>	45.78
Photo	43.82	<u>46.83</u>	47.19
Posing	37.03	37.34	<u>37.04</u>
Purchases	36.57	<u>37.16</u>	38.44
Sitting	56.83	<u>57.82</u>	59.29
SittingDown	<u>57.21</u>	57.09	64.64
Smoking	41.98	<u>42.87</u>	42.99
Waiting	38.95	37.92	<u>38.34</u>
WalkDog	40.17	<u>41.63</u>	43.76
Walking	31.1	<u>32.17</u>	32.98
WalkTogether	32.23	<u>33.67</u>	37.23
Average	40.64	<u>41.59</u>	43

model complexity as we add another CHI layer which increases interaction between hypotheses.

6.3. Qualitative Results

We report the qualitative results for transformer-based model experiments in Figure 5, 6, and 7. Inference is done

on all three models we have trained, with the input of a wild video of a dancing girl. We have randomly selected three challenging frames that some body parts are occluded, and compare the 3D pose estimation results from all three models. Arrow labels are set on the estimated 3D pose visualization to show the difference between the outputs, with the green color showing the best, the orange color showing the intermediate and the red showing the least ideal performance.

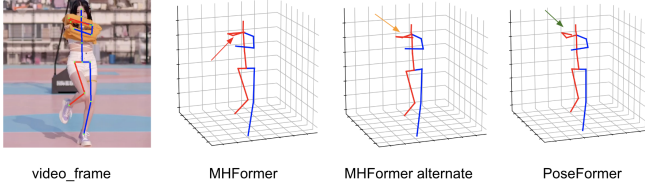


Figure 5. Visualization 1

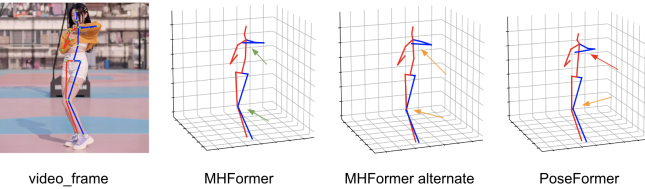


Figure 6. Visualization 2

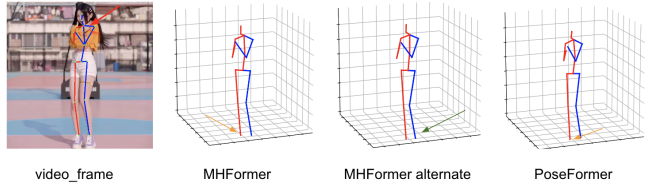


Figure 7. Visualization 3

For visualization 1, PoseFormer outperforms MHFormer Original and the MHFormer Alt because it correctly captures the position of the right arm that is tilting down. For visualization 2, MHFormer Original outperforms the others because it correctly captures the relative height between the left arm and the neck, as well as the intersection of two knees. For visualization 3, the MHFormer Alt outperforms the others because it correctly captures the front and back position of two feet. We also observe that in Visualization 3, the 2D keypoints of the right hand are captured incorrectly. From the qualitative results, all three models that we are testing have comparable performance, and there is no solid evidence showing which transformer model is the best.

7. Discussion

There are a number of interesting results in the quantitative data about MPJPE values that are reasonable, with respect to occlusion, the action involved and how each the transformer network works. Actions such as discussion, greetings and phoning, involve overlap of the hands when observing the input videos from the dataset, and hence have higher level of occlusion. In these types of actions, we observe MHFormer performing better than our MHF-Alt and PoseFormer in MPJPE because MHFormer generates multiple hypotheses. The similar observation is found in Sitting and SittingDown, where heavy occlusion with horizontal 2D joints formation and evinces the power of MHFormer over our model and the PoseFormer.

There are some types of actions that we observed a better performance in PoseFormer, such as posing and photo where there are less occlusions. This phenomenon can be observed in Fig 5 where PoseFormer performs better when the person has both hands shown towards the camera. This is because single hypothesis is capable of predicting actions that involve less occlusions, while multi-hypothesis adds too much variance.

We also observe that in situations where there is heavy occlusions such as sitting and sittingDown, the average MPJPE values rise significantly as compared to the other actions. It is concluded that occlusions have a key effect on the model performance.

In addition, while experimenting with videos in-the-wild, which consists of human motion of only top portion of the body, we were not able to obtain the 2D detection and hence no 3D HPE is generated. We found that MHFormer method is root-relative method and hence the video frames needs at least the root location to predict the 3D HPE. This is one of the important limitation when considering the application for the HPE using MHFormer method.

It is better to train on 2D ground truth data instead of 2D predictions obtained from CPN network and this can be clearly observed in the Table 3. It evinces ground truth data has lower MPJPE values in the different actions as compared to the other two. Observing from Table 3, CPN output was less accurate as compared with stacked glass 2D generated points.

8. Conclusions & Future Work

From the results and discussion, we conclude that MHFormer performs better on the human actions when there are heavy occlusions and PoseFormer performs better when there are less occlusions. From our experiments, we conclude that SHR, CHI arrangement (4 (b)) does not perform at par with the MHFormer paper [13]. However, a better architecture potentially exists which can outperform MHFormer.

Table 3. Impact of different inputs on MPJPE results

Actions	GT-PF	CPN-PF	Stacked Glass-Baseline
Directions	39.07	48.77	<u>42.86</u>
Discussion	<u>49.15</u>	59.05	47.91
Eating	44.34	<u>44.98</u>	48.92
Greeting	43.9	54.41	<u>49.64</u>
Phoning	42.56	61.39	<u>54.46</u>
Photo	39.77	<u>61.14</u>	65.28
Posing	44.0	47.5	<u>47.49</u>
Purchases	42.67	50.25	<u>48.88</u>
Sitting	<u>60.29</u>	76.21	59.56
SittingDown	55.29	85.07	<u>68.69</u>
Smoking	46.76	56.36	<u>51.68</u>
Waiting	44.06	<u>51.02</u>	51.58
WalkDog	39.8	<u>54.39</u>	55.38
Walking	37.21	39.85	<u>42.69</u>
WalkTogether	34.73	43.98	46.45
Average	44.2	55.59	<u>52.1</u>

There are several directions for further experiments. First, we can train more epochs until convergence for all the three models. Second, we can experiment other changes on the MHFormer architecture based on the results of our MHFormer Alt, for example, changing the proportionality of the SHR to CHI. Third, since the self-attention transformer structure in the MHFormer suffers from the discontinuity of the batches, transformer-XL could be fused into the MHFormer architecture so as long-term temporal dependency can be realized.

Following the conclusion and the potential experiments mentioned above, a unique architecture that combines MHFormer, PoseFormer and Transformer-XL could be proposed to increase the accuracy and push the boundaries of lifting 2D to 3D HPE.

Table 4. Training Validation Performance for 15 Epochs

Method: Baseline 3D		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Epoch		9.90E-04	9.80E-04	9.71E-04	9.61E-04	9.51E-04	9.42E-04	9.33E-04	9.23E-04	9.14E-04	9.05E-04	8.96E-04	8.87E-04	8.79E-04	8.71E-04	8.65E-04
Learning rate:		0.1193	0.0783	0.0681	0.0631	0.0598	0.0576	0.0558	0.0543	0.0532	0.0523	0.0514	0.0508	0.0501	0.0496	0.0491
Train loss avg:																
Action																
Directions	45.24	42.86	40.63	42.33	39.99	42.34	42.44	42.13	42.13	42.99	40.5	46.77	43.81	40.12	14	15
Discussion	50.52	47.91	47	46.7	46.21	47.9	45.23	47.13	47.13	49.04	48.05	50.12	47.56	45.13	42.7	41.72
Eating	51.62	48.92	45.89	47.85	44.58	46.66	44.91	46.01	46.01	45.47	44.31	47.12	47.53	43.23	45.7	45.49
Greeting	52.05	49.64	45.99	47.28	45.13	46.73	47.52	46.97	46.97	48.03	45.16	50.35	47.94	44.14	48.4	44.39
Phoning	58.82	54.46	52.34	52.53	51.02	50.76	51.51	50.42	50.42	50.9	50.87	52.5	51.34	48.3	47.59	46.4
Photo	70.96	65.28	61.32	61.88	58.53	59.77	60.26	59.57	59.57	61.37	56.84	59.4	61.49	57.18	50.76	48.7
Posing	50.26	47.49	46.26	46.49	45.83	46.93	48.72	47.61	47.61	50.32	46.97	52.03	48.99	45.61	59.65	57.81
Purchases	51.71	48.88	44.84	46.36	45.32	45.04	45.11	45.39	45.39	45.21	43.33	48.05	45.51	43.87	47.71	47.05
Sitting	63.48	59.56	55.94	57.41	55.61	56.04	56.71	56.27	56.27	58.41	58.48	58.29	56.3	54.82	45.25	43.06
SittingDown	74.94	68.69	65.74	66.77	66.04	63.9	62.71	61.96	61.96	65.97	65.77	63.26	62.05	63.09	55.72	54.4
Smoking	54.26	51.68	49.01	49.97	48.28	48.81	48.73	48.29	48.29	49.34	48.62	50.96	49.03	46.02	60.3	59.37
Waiting	55.67	51.58	49.76	49.4	48.6	48.54	49.01	48.97	48.97	50.38	48.46	51.86	50.24	46.85	48.46	46.46
WalkDog	60.15	55.38	52.87	53.17	50.66	52.6	50.47	52.2	52.2	52.45	51.04	54.95	53.08	50.1	48.23	47.39
Walking	47.19	42.69	40.24	43.02	39.77	40.18	41.24	41.81	41.81	40.38	39.17	45.25	42.85	38.93	52.32	51
WalkTogether	49.81	46.45	44.34	46.99	43.5	46.63	46.48	45.73	45.73	44.28	44.89	51.14	46.57	43.03	42.45	40.4
Average	55.78	52.1	49.48	50.54	48.6	49.52	49.4	49.36	49.36	50.3	48.83	52.14	50.29	47.36	47.43	44.18

9. Contributions

MHFormer github repository:

<https://github.com/Vegetebird/MHFormer>

PoseFormer github repository:

<https://github.com/zczcwh/PoseFormer>

Jiwen Chen: Train on the PoseFormer from scratch, and inference the visualization of PoseFormer on a in-wild video. Literature review, architecture design, and report writing.

Amoh Benjamin: Trained the MHFormer from scratch and run inference and visualization on the MHFormer architecture. Literature review on MHFormer, PoseFormer, Transformer-XL and other architecture, report writing and editing and architecture modeling.

Vasu Patel has worked on training the baseline model, MHFormer alternate, obtaining the 2D keypoint detections of Human 3.6M. Inferencing and generating visualizations for MHFormer alternate method. Perusing research papers on MHFormer, PoseFormers and discussing with the group about potential changes to the model architecture. Report writing, editing.

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